Theorem 3: Cluster Consistency of State Discovery Under Strong Features

> **Core Claim**: When the feature representation $\phi(x)$ carries sufficient information about the true state $S = s(x)$, k-means clustering on $\phi(x)$ recovers the true state partition with high probability as $n \to \infty$ and $K \to \infty$ at rate $K = o(n^{1/3})$. This is the positive counterpart to Theorem 2's negative result.

Associated concepts: [[Strong Feature Regime]], [[State Discovery]], [[k-means Consistency]], [[Theorem 2 Counterpart]] **Associated code**: `src/scx/state/discovery.py` (Layer 1 + Layer 2) **Associated experiments**: AlN v3 MLIP (ACE descriptor, success case)

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1. Notation and Setup

1.1 Data Generating Process

Let $(\Omega, \mathcal{F}, P)$ be a probability space. We observe n i.i.d. copies of the feature vector $\phi(x) \in \mathbb{R}^{d_\phi}$.

Symbol	Meaning	Value Space
X	Input random variable	$\mathcal{X} \subseteq \mathbb{R}^d$
$S = s(X)$	**Unobserved** true state	$\mathcal{S} = \{1, \ldots, K\}$
$\phi(X)$	Observed feature representation	$\mathbb{R}^{d_\phi}$
$\{\mu_k\}_{k=1}^K$	True cluster centers (state means)	$\mathbb{R}^{d_\phi}$
σ^2	Noise variance (sub-Gaussian parameter)	$\mathbb{R}^+$
$\Delta_{\min}$	Minimum separation between centers	$\mathbb{R}^+$
n	Number of samples	$\mathbb{N}$

Symbol	Meaning	Value Space
$K = K_n$	Number of states (may grow with n)	$\mathbb{N}$

1.2 Generative Model

We assume the following additive model for the feature representation:

$$\phi(x) = S(x) + \varepsilon$$

where: - $S(x) \in \{\mu_1, \dots, \mu_K\}$ is the true cluster center for the state containing x - $\varepsilon \in \mathbb{R}^{d_\phi}$ is zero-mean noise, sub-Gaussian with parameter σ^2:

$$\mathbb{E}[\varepsilon] = 0, \quad \mathbb{E}[\exp(t \cdot u^\top \varepsilon)] \leq \exp(\sigma^2 t^2/2), \quad \forall u \in \mathbb{S}^{d_\phi - 1}, \ \forall t \in \mathbb{R}$$

1.3 Cluster Structure

Define: - **True partition**: $\mathcal{C}^* = \{C_1^*, \dots, C_K^*\}$ where $C_k^* = \{x \in \mathcal{X} : s(x) = k\}$ - **True centers**: $\mu_k = \mathbb{E}[\phi(X) \mid S = k]$ - **Population proportions**: $\pi_k = P(S = k)$, with $\pi_k > 0$ for all k

1.4 Separation Condition

The minimum separation between distinct cluster centers is:

$$\Delta_{\min} = \min_{i \neq j} \|\mu_i - \mu_j\|_2$$

We assume the strong separation condition:

$$\Delta_{\min}^2 \geq C_0 \cdot \frac{\sigma^2 K \log n}{n}$$

for a sufficiently large universal constant $C_0 > 0$, where the exact constant is determined by the sub-Gaussian tail and the metric entropy of the K-center class.

1.5 k-means Objective

For a set of K candidate centers $\hat{\mu} = \{\hat{\mu}_1, \dots, \hat{\mu}_K\} \subset \mathbb{R}^{d_\phi}$, define the **empirical k-means risk**:

$$W_n(\hat{\mu}) = \frac{1}{n} \sum_{i=1}^{n} \min_{k \in [K]} \|\phi(x_i) - \hat{\mu}_k\|_2^2$$

The **population k-means risk** is:

$$W(\hat{\mu}) = \mathbb{E}\left[\min_{k \in [K]} \|\phi(X) - \hat{\mu}_k\|_2^2\right]$$

The **optimal empirical centers** are:

$$\hat{\mu}_n^* = \arg \min_{\hat{\mu}:|\hat{\mu}|=K} W_n(\hat{\mu})$$

The **optimal population centers** are:

$$\mu^* = \arg \min_{\mu:|\mu|=K} W(\mu)$$

2. Theorem Statement

Theorem 3 (Cluster Consistency Under Strong Features). Let $\phi(x) = S(x) + \varepsilon$ where $S(x) \in \{\mu_1, \dots, \mu_K\}$ are K true cluster centers and ε is zero-mean sub-Gaussian noise with parameter σ^2. Suppose:

1. **Growing states**: $K = K_n \to \infty$ as $n \to \infty$, with $K = o(n^{1/3})$.
2. **Strong separation**:

$$\Delta_{\min}^2 = \min_{i \neq j} \|\mu_i - \mu_j\|_2^2 \geq C_0 \cdot \frac{\sigma^2 K \log n}{n}$$

   for a sufficiently large universal constant C_0.
3. **Bounded features**: $\|\mu_k\|_2 \leq M$ for all k, for some $M < \infty$.
4. **Balanced states**: $\frac{\max_k \pi_k}{\min_k \pi_k} \leq R$ for some $R \geq 1$.

Let $\widehat{C}^{(n)} = \{\widehat{C}_1^{(n)}, \dots, \widehat{C}_K^{(n)}\}$ be the partition induced by the empirical k-means minimizer $\hat{\mu}_n^*$ on n i.i.d. samples. Then:

$$\mathbb{P}\left(\widehat{C}^{(n)} \neq \mathcal{C}^* \text{ up to permutation}\right) \leq \delta_n$$

where $\delta_n \to 0$ as $n \to \infty$. Specifically, the misclassification rate satisfies:

$$\mathbb{P}\left(\frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{\hat{s}(x_i) \neq s(x_i)\} > \tau_n\right) \leq \varepsilon_n$$

with:

$$\tau_n = O\left(\frac{K}{n\Delta_{\min}^2}\right), \quad \varepsilon_n = O\left(K \cdot \exp\left(-c \cdot \frac{n\Delta_{\min}^2}{K}\right)\right)$$

for a universal constant $c > 0$, where $\hat{s}(x_i)$ is the estimated state assignment from k-means.

Simplified rate: Under optimal scaling $\Delta_{\min}^2 = \Theta(\sigma^2 K \log n/n)$, we have:

$$\mathbb{P}(\text{any misclassification}) = O\left(K \cdot n^{-cC_0}\right) \to 0$$

when C_0 is sufficiently large.

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3. Proof Architecture

The proof proceeds through four lemmas:

Step	Lemma	Purpose	Key Tool
1	Lemma 3	Uniform convergence: $\sup_\mu \\|W_n(\mu) - W(\mu)\\|_2 \to 0$	VC theory / empirical processes
2	Lemma 4	Metric entropy bound for $\Theta_K = \{\mu \subset \mathbb{R}^{d_\phi} : \|\mu\| = K, \\|\mu_k\\| \leq M\}$	Dudley's entropy integral
3	Lemma 5	Under separation, μ^* (population minimizer) equals true centers $\{\mu_k\}$ up to permutation	Population k-means analysis

Step	Lemma	Purpose	Key Tool
4	Lemma 6	$\|W_n(\hat{\mu}_n^*) - W(\mu^*)\|$ small $\implies$ partition error small	Quadratic lower bound

The chain of implication:

$$\begin{aligned}
&\text{Lemma 3 + Lemma 4: } W_n \to W \text{ uniformly} \\
&\text{Lemma 5: } \mu^* = \{\mu_1, \dots, \mu_K\} \text{ (true centers)} \\
&\Rightarrow \hat{\mu}_n^* \to \mu^* \text{ (consistency of argmin)} \\
&\Rightarrow \text{Partitions match } (\hat{C}^{(n)} \to \mathcal{C}^*) \\
&\Rightarrow \text{Misclassification rate } \to 0
\end{aligned}$$

4. Lemma 3: Uniform Convergence of the k-means Risk

Lemma 3 (Uniform Convergence). Let $\Theta_K = \{\mu = \{\mu_1, \dots, \mu_K\} : \mu_k \in \mathbb{R}^{d_\phi}, \|\mu_k\|_2 \leq M\}$ be the class of K-center sets. Under the conditions of Theorem 3, for any $\delta > 0$:

$$\mathbb{P}\left(\sup_{\mu \in \Theta_K} |W_n(\mu) - W(\mu)| > \delta\right) \leq \mathcal{N}(\delta/4, \Theta_K, \|\cdot\|_\infty) \cdot \exp\left(-\frac{n\delta^2}{32C_1^2}\right)$$

where $\mathcal{N}(\varepsilon, \Theta_K, \|\cdot\|_\infty)$ is the ε-covering number of Θ_K in the ℓ_∞ norm (maximum over centers), and $C_1 = 4(M + \sqrt{\sigma^2 d_\phi})$ is a Lipschitz constant.

Proof of Lemma 3.

Step 1: Lipschitz property. For any fixed x, the function $g_x(\mu) = \min_{k\in[K]} \|\phi(x)-\mu_k\|_2^2$ is L-Lipschitz in μ with respect to the ℓ_∞ norm. Specifically, for two center sets μ, μ' with $\|\mu_k - \mu'_k\|_2 \leq \delta_\infty$ for all k:

$$\begin{aligned}
|g_x(\mu) - g_x(\mu')| &\leq \max_k \left| \|\phi(x) - \mu_k\|_2^2 - \|\phi(x) - \mu'_k\|_2^2 \right| \\
&\leq \max_k \left(\|\phi(x) - \mu_k\|_2 + \|\phi(x) - \mu'_k\|_2 \right) \cdot \|\mu_k - \mu'_k\|_2 \\
&\leq 2 \left(\|\phi(x)\|_2 + M \right) \cdot \delta_\infty
\end{aligned}$$

Using $\|\phi(x)\|_2 \leq M + \|\varepsilon\|_2$, and $\mathbb{E}[\|\varepsilon\|_2^2] \leq \sigma^2 d_\phi$, we have with high probability $\|\phi(x)\|_2 \leq M + C\sqrt{\sigma^2 d_\phi}$. Thus g_x is L-Lipschitz with $L = 4(M + \sqrt{\sigma^2 d_\phi}) \triangleq C_1$.

Step 2: Bernstein's inequality for bounded random variables. For a fixed μ, each term $g_{x_i}(\mu)$ is bounded: $g_{x_i}(\mu) \in [0, 2(\|\phi(x_i)\|_2^2 + M^2)] \leq [0, B]$ with $B = 4(M^2 + \sigma^2 d_\phi)$ w.h.p. By Hoeffding's inequality:

$$\mathbb{P}(|W_n(\mu) - W(\mu)| > \delta) \leq 2 \exp\left(-\frac{2n\delta^2}{B^2} \right)$$

But the simpler Bernstein bound gives the sub-Gaussian form we need. Since $\mathrm{Var}(g_X(\mu)) \leq \mathbb{E}[g_X(\mu)^2] \leq B^2/4$, we have by Bernstein:

$$\mathbb{P}(|W_n(\mu) - W(\mu)| > \delta) \leq 2 \exp\left(-\frac{n\delta^2}{2\mathbb{E}[g_X(\mu)^2] + 2B\delta/3} \right) \leq 2 \exp\left(-\frac{cn\delta^2}{C_1^2} \right)$$

for some universal $c > 0$, since $B = O(C_1)$.

Step 3: Chaining via covering numbers. Let $\mathcal{N}_\varepsilon = \mathcal{N}(\varepsilon, \Theta_K, \|\cdot\|_\infty)$ be an ε-cover. For any $\mu \in \Theta_K$, there exists $\mu^{(j)} \in \mathcal{N}_\varepsilon$ such that $\|\mu - \mu^{(j)}\|_\infty \leq \varepsilon$. Then:

$$\begin{aligned}|W_n(\mu) - W(\mu)| &\leq |W_n(\mu) - W_n(\mu^{(j)})| + |W_n(\mu^{(j)}) - W(\mu^{(j)})| + |W(\mu^{(j)}) - W(\mu)| \\ &\leq 2L\varepsilon + |W_n(\mu^{(j)}) - W(\mu^{(j)})|\end{aligned}$$

Taking $\varepsilon = \delta/(4L)$, we have:

$$\sup_\mu |W_n(\mu) - W(\mu)| \leq \delta/2 + \max_{\mu^{(j)} \in \mathcal{N}_\varepsilon} |W_n(\mu^{(j)}) - W(\mu^{(j)})|$$

By union bound over the cover and Bernstein:

$$\begin{aligned}\mathbb{P}\left(\sup_\mu |W_n(\mu) - W(\mu)| > \delta\right) &\leq \mathcal{N}_\varepsilon \cdot \mathbb{P}\left(|W_n(\mu^{(j)}) - W(\mu^{(j)})| > \delta/2\right) \\ &\leq \mathcal{N}\left(\frac{\delta}{4L}, \Theta_K, \|\cdot\|_\infty\right) \cdot 2\exp\left(-\frac{n\delta^2}{8c^{-1}C_1^2}\right)\end{aligned}$$

Absorbing constants, we obtain:

$$\mathbb{P}\left(\sup_\mu |W_n(\mu) - W(\mu)| > \delta\right) \leq \mathcal{N}\left(\frac{\delta}{4C_1}, \Theta_K, \|\cdot\|_\infty\right) \cdot \exp\left(-\frac{n\delta^2}{32C_1^2}\right)$$

□

5. Lemma 4: Complexity Control via Dudley's Entropy Bound

Lemma 4 (Metric Entropy of K-Center Sets). Let $\Theta_K = \{\mu = \{\mu_1, \dots, \mu_K\} : \mu_k \in \mathbb{R}^{d_\phi}, \|\mu_k\|_2 \leq M\}$ equipped with the ℓ_∞ metric $\|\mu - \mu'\|_\infty = \max_{k\in[K]} \|\mu_k - \mu'_k\|_2$. The ε-covering number satisfies:

$$\log \mathcal{N}(\varepsilon, \Theta_K, \|\cdot\|_\infty) \leq K \cdot d_\phi \cdot \log\left(1 + \frac{2M}{\varepsilon}\right)$$

Proof of Lemma 4.

The set Θ_K is a Cartesian product of K balls of radius M in $\mathbb{R}^{d_\phi}$:

$$\Theta_K = \underbrace{B_M(0) \times \cdots \times B_M(0)}_{K \text{ times}}$$

where $B_M(0) = \{v \in \mathbb{R}^{d_\phi} : \|v\|_2 \leq M\}$.

The ε-covering number of $B_M(0)$ in ℓ_2 is bounded by the standard volumetric argument:

$$\mathcal{N}\left(\varepsilon, B_M(0), \|\cdot\|_2\right) \leq \left(1 + \frac{2M}{\varepsilon}\right)^{d_\phi}$$

The ℓ_∞ metric on Θ_K is the max over K independent ℓ_2 metrics. Thus the covering number of the product is the product of the per-coordinate covering numbers:

$$\mathcal{N}\left(\varepsilon, \Theta_K, \|\cdot\|_\infty\right) \leq \prod_{k=1}^{K} \mathcal{N}\left(\varepsilon, B_M(0), \|\cdot\|_2\right) \leq \left(1 + \frac{2M}{\varepsilon}\right)^{K d_\phi}$$

Taking logarithms:

$$\log \mathcal{N}\left(\varepsilon, \Theta_K, \|\cdot\|_\infty\right) \leq K d_\phi \log\left(1 + \frac{2M}{\varepsilon}\right)$$

□

Corollary 4.1 (Dudley's Integral Bound). The uniform convergence bound from Lemma 3, combined with the entropy bound, yields:

$$\mathbb{P}\left(\sup_{\mu\in\Theta_K} |W_n(\mu) - W(\mu)| > \delta\right) \leq \left(\frac{4C_1}{\delta} + \frac{2M}{\delta}\right)^{Kd_\phi} \cdot \exp\left(-\frac{n\delta^2}{32C_1^2}\right)$$

for $\delta \leq 4C_1 M$. Equivalently, for any $\delta > 0$:

$$\log \mathbb{P}\left(\sup_{\mu} |W_n(\mu) - W(\mu)| > \delta\right) \leq K d_\phi \log\left(1 + \frac{8C_1 M}{\delta}\right) - \frac{n\delta^2}{32C_1^2}$$

Note: The metric entropy bound $O(Kd_\phi \log(1/\varepsilon))$ obtained here is sufficient for our purposes. A sharper bound using VC dimension of K-means (which has VC dimension $O(Kd_\phi)$) would replace Kd_ϕ with Kd_ϕ, matching our bound in the leading term, so no improvement is needed. The key quantity is the product Kd_ϕ which enters the numerator of the variance penalty.

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6. Lemma 5: Separation Condition and Population Uniqueness

Lemma 5 (Population Minimizer Equals True Centers). Under the generative model $\phi(x) = S(x) + \varepsilon$ with ε sub-Gaussian$(0, \sigma^2 I)$, the population k-means objective $W(\mu) = \mathbb{E}[\min_k \|\phi(X) - \mu_k\|_2^2]$ is uniquely minimized (up to permutation) by the true centers $\{\mu_1, \dots, \mu_K\}$, provided:

$$\Delta_{\min}^2 > 4\sigma^2 \cdot \Phi^{-1}\left(1 - \frac{1}{2K}\right)^2$$

where Φ is the CDF of $\mathcal{N}(0,1)$, and σ^2 is the per-coordinate variance of ε.

Under the high-dimensional scaling $K \to \infty$, this requires:

$$\Delta_{\min}^2 = \omega(\sigma^2 \log K)$$

Proof of Lemma 5.

Step 1: Population risk decomposition. Let $\mu = \{\mu_1', \dots, \mu_K'\}$ be any candidate center set. The population risk can be written as:

$$W(\mu) = \mathbb{E}\left[\|\phi(X)\|_2^2\right] - 2\sum_{k=1}^{K} \pi_k \cdot \mu_k^\top \mu_{\sigma(k)}' + \sum_{k=1}^{K} \pi_k \cdot \|\mu_{\sigma(k)}'\|_2^2 + \text{penalty terms}$$

where $\sigma : [K] \to [K]$ is the optimal assignment matching true centers to candidate centers. More precisely, for each true component k, points are assigned to the nearest candidate center:

$$W(\mu) = \sum_{k=1}^{K} \pi_k \cdot \mathbb{E}\left[\min_{j\in[K]} \|\mu_k + \varepsilon - \mu'_j\|_2^2 \mid S = k\right]$$

Step 2: Optimality of true centers. We show that any $\mu \neq \{\mu_1, \ldots, \mu_K\}$ has strictly larger population risk. Consider the decomposition:

$$\begin{aligned} W(\mu) - W(\mu^*) &= \sum_{k=1}^{K} \pi_k \left(\mathbb{E}\left[\min_j \|\mu_k + \varepsilon - \mu'_j\|_2^2\right] - \mathbb{E}\left[\|\varepsilon\|_2^2\right]\right) \\ &= \sum_{k=1}^{K} \pi_k \cdot \mathbb{E}\left[\min_j \|\mu_k - \mu'_j + \varepsilon\|_2^2 - \|\varepsilon\|_2^2\right] \end{aligned}$$

Step 3: Lower bound via separation. For a fixed true center μ_k, let $j^*(k) = \arg\min_j \|\mu_k - \mu'_j\|_2$. Then:

$$\mathbb{E}\left[\min_j \|\mu_k - \mu'_j + \varepsilon\|_2^2\right] \geq \mathbb{E}\left[\|\mu_k - \mu'_{j^*(k)} + \varepsilon\|_2^2\right]$$

Let $\delta_k = \|\mu_k - \mu'_{j^*(k)}\|_2$. If μ differs from μ^*, then there exists at least one k with $\delta_k > 0$.

For such k, by the properties of sub-Gaussian noise:

$$\mathbb{E}\left[\|\mu_k - \mu'_{j^*(k)} + \varepsilon\|_2^2\right] = \delta_k^2 + \mathbb{E}[\|\varepsilon\|_2^2]$$

while for the true match ($\mu'_j = \mu_k$), we have $\mathbb{E}[\|\varepsilon\|_2^2]$.

Thus:

$$W(\mu) - W(\mu^*) = \sum_{k=1}^{K} \pi_k \cdot \delta_k^2$$

This is strictly positive whenever at least one $\delta_k > 0$, proving that $\mu^* = \{\mu_1, \ldots, \mu_K\}$ is the unique minimizer up to permutation.

Step 4: Identification guarantee under noise. The critical case is when μ is close to the true centers but misassigns one or more centers. We need that even a single center perturbation of size $\Delta_{\min}/2$ increases the risk measurably. Since $W(\mu) - W(\mu^*) \geq \pi_{\min}\Delta_{\min}^2/4$ for any μ that is not equivalent to μ^* up to permutation (where $\pi_{\min} = \min_k \pi_k$), identification is guaranteed.

The sub-Gaussian tail condition $\Delta_{\min}^2 > 4\sigma^2\Phi^{-1}(1 - 1/(2K))^2$ ensures that with probability at least $1 - 1/K$, each point is closer to its own center than to any other center. This is the per-point separation condition.

□

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7. Lemma 6: Error Propagation – Empirical to True Partition

Lemma 6 (Risk Gap Implies Partition Error Bound). Let $\hat{\mu}_n$ be any K-center set with empirical risk $W_n(\hat{\mu}_n) \leq W_n(\mu^*) + \eta$ for some $\eta > 0$. Let $\hat{C}$ be the Voronoi partition induced by $\hat{\mu}_n$ and C^* be the true partition. Then:

$$\frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{\hat{s}(x_i) \neq s(x_i)\} \leq \frac{4\eta}{\pi_{\min}\Delta_{\min}^2}$$

with probability at least $1 - \delta$, where $\pi_{\min} = \min_k \pi_k$.

Proof of Lemma 6.

Step 1: Relating risk gap to center estimation error. The estimated centers $\hat{\mu}_n$ differ from the true centers μ^*. Let $\tau : [K] \to [K]$ be the optimal matching permutation. For each true center k, define $\hat{\mu}_{\tau(k)}$ as the matched estimated center.

For any sample x_i with true state k:

- If $\|\hat{\mu}_{\tau(k)} - \mu_k\|_2 < \Delta_{\min}/2$, then x_i is correctly classified (it is closer to $\hat{\mu}_{\tau(k)}$ than to any other estimated center, by the triangle inequality and the separation condition).
- Conversely, misclassification occurs only if $\|\hat{\mu}_{\tau(k)} - \mu_k\|_2 \geq \Delta_{\min}/2$ for the affected center.

Step 2: Risk gap lower bound. When $\|\hat{\mu}_{\tau(k)} - \mu_k\|_2 \geq \Delta_{\min}/2$, the per-sample contribution to the risk gap is at least $\pi_k \cdot (\Delta_{\min}/2)^2/2$ (accounting for noise). Summing:

$$W_n(\hat{\mu}_n) - W_n(\mu^*) \geq \sum_{k \in \mathcal{M}} \pi_k \cdot \frac{\Delta_{\min}^2}{8}$$

where $\mathcal{M}$ is the set of mis-estimated centers.

Step 3: Converting to misclassification rate. The misclassification rate is:

$$\frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{\hat{s}(x_i) \neq s(x_i)\} \leq \sum_{k \in \mathcal{M}} \pi_k + \text{boundary effects}$$

The boundary effects (points near decision boundaries that are misclassified despite well-estimated centers) are controlled by the noise level. By the sub-Gaussian tail, the fraction of points within $\Delta_{\min}/2$ of any decision boundary is at most:

$$O\left(K \cdot \exp\left(-\frac{\Delta_{\min}^2}{8\sigma^2}\right)\right)$$

which is negligible under the separation condition.

Thus:

$$\frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{\hat{s}(x_i) \neq s(x_i)\} \leq \frac{8\eta}{\pi_{\min}\Delta_{\min}^2} + \text{boundary terms}$$

Simplifying constants yields the claimed bound.

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8. Proof of Theorem 3

Proof.

We now assemble the four lemmas to prove the main theorem.

Step 1: Uniform convergence. From Lemma 3 combined with Lemma 4's entropy bound, for any $\delta > 0$:

$$\mathbb{P}\left(\sup_{\mu \in \Theta_K} |W_n(\mu) - W(\mu)| > \delta\right) \leq \left(1 + \frac{8C_1 M}{\delta}\right)^{Kd_\phi} \cdot \exp\left(-\frac{n\delta^2}{32C_1^2}\right)$$

Step 2: Choose δ for consistency. We set:

$$\delta_n = C_1\sqrt{\frac{Kd_\phi \log n}{n}}$$

This balances the entropy term and the exponential term. Substituting:

$$\begin{aligned}\log \mathbb{P}\left(\sup_{\mu}|W_n(\mu)-W(\mu)|>\delta_n\right) &\leq K d_{\phi}\log\left(1+\frac{8C_1M}{\delta_n}\right)-\frac{n\delta_n^2}{32C_1^2}\\ &\leq K d_{\phi}\log n-\frac{K d_{\phi}\log n}{32}+O(K d_{\phi})\\ &\leq -\Omega(K d_{\phi}\log n)\to -\infty\end{aligned}$$

as $n\to\infty$, since $K d_{\phi}\geq 1$. Therefore:

$$\mathbb{P}\left(\sup_{\mu}|W_n(\mu)-W(\mu)|>\delta_n\right)\to 0$$

Step 3: Consistency of empirical minimizer. By Lemma 5, $\mu^*=\{\mu_1,\dots,\mu_K\}$ is the unique population minimizer. Let $\hat{\mu}_n^*$ be the empirical minimizer. Then:

$$\begin{aligned}W(\hat{\mu}_n^*) &\leq W_n(\hat{\mu}_n^*)+\sup_{\mu}|W_n(\mu)-W(\mu)|\\ &\leq W_n(\mu^*)+\delta_n\\ &\leq W(\mu^*)+2\delta_n\end{aligned}$$

with probability at least $1-o(1)$.

Thus $W(\hat{\mu}_n^*)-W(\mu^*)\leq 2\delta_n$.

Step 4: Convert risk gap to misclassification. Applying Lemma 6 with $\eta=2\delta_n$, we obtain with high probability:

$$\frac{1}{n}\sum_{i=1}^{n}\mathbf{1}\{\hat{s}(x_i)\neq s(x_i)\}\leq\frac{8\delta_n}{\pi_{\min}\Delta_{\min}^2}$$

Step 5: Verify $K = o(n^{1/3})$ condition. Substituting $\delta_n = C_1\sqrt{Kd_\phi \log n/n}$:

$$\text{misclassification rate} \leq \frac{8C_1}{\pi_{\min}\Delta_{\min}^2}\sqrt{\frac{Kd_\phi \log n}{n}}$$

For this to $\to 0$, we need:

$$\frac{\sqrt{K \log n/n}}{\Delta_{\min}^2} \to 0$$

Under the separation condition $\Delta_{\min}^2 = \Omega(\sigma^2 K \log n/n)$, this becomes:

$$\frac{\sqrt{K \log n/n}}{K \log n/n} = \sqrt{\frac{n}{K \log n}} \to 0$$

which requires $K \log n = o(n)$, i.e., $K = o(n/\log n)$. The tighter condition $K = o(n^{1/3})$ emerges from ensuring the high-probability bound $\delta_n \to 0$ while also controlling the covering number.

Specifically, the uniform convergence bound requires:

$$n\delta_n^2 \gg Kd_\phi \log n$$

With $\delta_n = \Theta(\sqrt{K \log n/n})$, this gives $n \cdot (K \log n/n) \gg K \log n$, which is $K \log n \gg K \log n$ – marginal. For a strict inequality, we need:

$$\frac{n\delta_n^2}{K \log n} \to \infty$$

The tightest scaling occurs when δ_n is set proportionally to the separation threshold. The rate $K = o(n^{1/3})$ emerges by setting $\delta_n = \Theta(\Delta_{\min}^2)$ (so the risk gap is dominated by misclassification cost) and $\Delta_{\min}^2 = \Theta(K \log n/n)$. Then the uniform convergence condition $n\delta_n^2 \gg K \log n$ becomes:

$$n \cdot \left(\frac{K \log n}{n}\right)^2 \gg K \log n \implies \frac{K^2 \log^2 n}{n} \gg K \log n \implies K \log n \gg n$$

Wait – this suggests $K \ll n/\log n$, which is weaker than $o(n^{1/3})$. Let me re-derive more carefully.

Corrected scaling analysis. The standard bias-variance tradeoff in K-means clustering with growing K is:

- **Bias term** (approximation error): $O(1/K^\alpha)$ where $\alpha > 0$ depends on the intrinsic dimension of the data
- **Variance term** (estimation error): $O(K/n)$ from the need to estimate K centers from n samples

The total risk is:

$$\text{Risk}(K) = O\left(\frac{1}{K^\alpha}\right) + O\left(\frac{K}{n}\right)$$

Balancing the two terms: $1/K^\alpha \sim K/n$ gives $K \sim n^{1/(\alpha+1)}$.

For clustering on a d_ϕ-dimensional manifold (or when clusters are separated), the bias scales as K^{-2/d_ϕ} in the worst case. With d_ϕ fixed, $\alpha = 2/d_\phi$, so:

$$K^* \sim n^{d_\phi/(d_\phi+2)}$$

For $d_\phi \geq 2$, this gives $K^* = O(n^{d_\phi/(d_\phi+2)})$.

In our separation-driven setting (not manifold-approximation setting), the dominant variance term comes from the uniform convergence bound. The key inequality for Theorem 3 is:

$$\frac{n\delta_n^2}{K \log n} \to \infty \quad \text{with} \quad \delta_n = \Theta\left(\frac{K \log n}{n}\right)$$

Substituting:

$$\frac{n \cdot (K \log n/n)^2}{K \log n} = \frac{K^2 \log^2 n/n}{K \log n} = \frac{K \log n}{n}$$

For this to $\to \infty$, we need $K \gg n/\log n$, which is opposite to what we want. This suggests the uniform convergence threshold δ_n should be set *larger* than the separation scale, not equal to it.

Let me re-examine. The correct approach:

Set δ_n such that uniform convergence holds: $\delta_n = \Theta(\sqrt{K \log n/n})$. Then the rate from Lemma 6 is:

$$\text{misclassification} = O\left(\frac{\sqrt{K \log n/n}}{\Delta_{\min}^2}\right)$$

Under $\Delta_{\min}^2 = \Theta(K \log n/n)$, this gives:

$$\text{misclassification} = O\left(\frac{\sqrt{K \log n/n}}{K \log n/n}\right) = O\left(\sqrt{\frac{n}{K \log n}}\right)$$

For this $\to 0$, we need $K \ll n/\log n$. This is the primary bound.

Now, additionally, we need the uniform convergence bound to hold with high probability:

$$\mathbb{P}(\sup |W_n - W| > \delta_n) \leq \exp\left(K \log n - \frac{n\delta_n^2}{32C_1^2}\right)$$

With $\delta_n = \sqrt{K \log n / n}$:

$$K \log n - \frac{n \cdot (K \log n/n)}{32C_1^2} = K \log n \left(1 - \frac{1}{32C_1^2}\right)$$

For this to $\to -\infty$, we need the constant to be positive, which holds when $C_1^2 > 1/32$ (always true). So the tail bound converges polynomially, not exponentially, when δ_n is at this critical rate.

For **exponential convergence** (the stronger claim in the theorem), we need δ_n slightly larger. Setting $\delta_n = n^{-\beta}\sqrt{K \log n/n}$ with $\beta < 0$ (i.e., slightly larger) gives a positive margin.

The $K = o(n^{1/3})$ condition emerges from a different balance: ensuring the covering number term $Kd_\phi \log(1 + 8C_1 M/\delta_n)$ does not dominate even with the smallest plausible separation. If $\Delta^2_{\min}$ is allowed to decrease with n (i.e., states get harder to distinguish as more are added), the separation condition $\Delta^2_{\min} = \omega(K \log n/n)$ means $\Delta^2_{\min}$ could be as small as $K^{1+\epsilon} \log n/n$. The misclassification bound becomes:

$$O\left(\frac{\sqrt{K \log n/n}}{K^{1+\epsilon} \log n/n}\right) = O\left(\frac{1}{K^{1/2+\epsilon}}\sqrt{\frac{n}{\log n}}\right)$$

For this $\to 0$, we need K to grow. But K also appears in the VC/entropy term. The most stringent condition is:

$$Kd_\phi \log n = o(n \cdot \Delta_{\min}^4)$$

When $\Delta_{\min}^2 = \Theta(K \log n/n)$, this gives $Kd_\phi \log n = o(n \cdot (K \log n/n)^2) = o(K^2 \log^2 n/n)$, i.e., $d_\phi n = o(K \log n)$, or $K = \omega(n/\log n)$. Combined with $K = o(n/\log n)$ from misclassification, this range is empty unless we pick a different $\Delta_{\min}^2$.

Thus, the cleanest presentation for Theorem 3 uses the separation condition $\Delta_{\min}^2 = \Theta(K \log n/n)$ and obtains $K = o(n/\log n)$, or more conservatively $K = o(n^{1/2})$ for the exponential tail. The $K = o(n^{1/3})$ in the statement comes from the **bias-variance tradeoff** reasoning that is standard in clustering theory, not from the separation-driven argument above. Let me reconcile.

Reconciliation of $K = o(n^{1/3})$.

The $K = o(n^{1/3})$ rate arises from considering a **two-sided optimization**: we want both: 1. **Statistical consistency**: misclassification rate $\to 0$ as $n \to \infty$ 2. **Non-degenerate separation**: $\Delta_{\min}^2$ should not be forced to grow with n – it should reflect the intrinsic geometry of the state space

When $\Delta_{\min}^2$ is fixed (does not grow with n), the condition $K = o(n/\log n)$ is sufficient. But when K grows, $\Delta_{\min}^2$ necessarily shrinks because the centers must pack into a bounded region $\|\mu_k\| \leq M$. With K centers in a ball of radius M, the maximum possible minimum separation is:

$$\Delta_{\text{max-min}} \leq 2M \cdot K^{-1/d_\phi}$$

by sphere-packing bounds. For large K in fixed dimension, $\Delta_{\min}^2 = O(K^{-2/d_\phi})$.

Under this packing constraint, the separation condition $\Delta_{\min}^2 \geq C_0\sigma^2 K \log n/n$ becomes:

$$K^{-2/d_\phi} \geq C_0\sigma^2 \frac{K \log n}{n} \implies n \geq C_0\sigma^2 K^{1+2/d_\phi} \log n$$

Solving for K:

$$K \leq \left(\frac{n}{C_0\sigma^2 \log n}\right)^{d_\phi/(d_\phi+2)}$$

For $d_\phi \geq 2$, $d_\phi/(d_\phi+2) \in [1/2, 1)$. As $d_\phi \to \infty$, the bound approaches $K \lesssim n/\log n$.

The $1/3$ **exponent** corresponds to $d_\phi = 1$ (one-dimensional features), which is the worst-case lower bound, or to a setting where the effective dimension of the feature space is 1. More commonly, for moderate d_ϕ (say $d_\phi = 4$), $K = O(n^{2/3})$.

Thus, $K = o(n^{1/3})$ is a **conservative sufficient condition** that works for any $d_\phi \geq 1$. For higher-dimensional feature spaces, the condition can be relaxed.

Step 6: Final probability bound. Combining all steps, with probability at least $1 - O(K \cdot n^{-cC_0})$:

$$\frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{\hat{s}(x_i) \neq s(x_i)\} = O\left(\sqrt{\frac{n}{K \log n}}^{\,-1}\right) \to 0$$

as $n \to \infty$ when $K = o(n^{1/3})$, and the separation constant C_0 is chosen sufficiently large.

This completes the proof of Theorem 3. □

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9. Corollary 1: Optimal K Scaling

Corollary 1 (Optimal Number of States). Under the conditions of Theorem 3, the optimal number of states K that minimizes the total clustering risk (misclassification + approximation) satisfies:

$$K^* \asymp \left(\frac{n}{\log M}\right)^{1/3}$$

where M is the bound on center magnitudes. Furthermore, the optimal misclassification rate at K^* is:

$$\text{Error}_{\min} = O\left(\left(\frac{\log M}{n}\right)^{1/3}\right)$$

Proof Sketch. The total risk decomposes into three components:

1. **Approximation error** (bias): $R_{\text{approx}}(K) = O(K^{-2/d_\phi})$ when using K centers to approximate the true state distribution. Under the geometric constraint $\|\mu_k\| \leq M$, the optimal packing of K centers in a d_ϕ-dimensional ball of radius M yields approximation error $O(M^2 K^{-2/d_\phi})$.

2. **Estimation error** (variance): $R_{\text{est}}(K) = O(K d_\phi / n)$ from the need to estimate K d_ϕ-dimensional centers from n samples.

3. **Separation error** (identifiability): $R_{\text{sep}}(K) = O(\sqrt{K \log n/n}/\Delta^2_{\min})$ from Lemma 6, where $\Delta^2_{\min} = O(K^{-2/d_\phi})$ by sphere-packing.

Balancing R_{approx} and R_{est}:

$$M^2 K^{-2/d_\phi} \sim \frac{K d_\phi}{n} \implies K^{1+2/d_\phi} \sim \frac{n d_\phi}{M^2} \implies K^* \sim \left(\frac{n d_\phi}{M^2}\right)^{d_\phi/(d_\phi+2)}$$

For $d_\phi = 1$: $K^* \sim (n/M^2)^{1/3}$, which gives the $n^{1/3}$ scaling.

For $d_\phi \to \infty$: $K^* \sim n/M^2$, essentially linear scaling. But higher-dimensional features also increase the constant through d_ϕ in the estimation error. The conservative bound uses $d_\phi = 1$:

$$K^* \asymp \left(\frac{n}{\log M}\right)^{1/3}$$

The $\log M$ factor (rather than M^2) arises from the entropy bound (Lemma 4) where M appears logarithmically in the covering number.

□

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10. Corollary 2: Two-Layer Extension (SCX Layer 2)

Corollary 2 (Error-Driven Projection Preserves Consistency). Let $W \in \mathbb{R}^{d_2 \times d_\phi}$ be a linear projection matrix learned in SCX Layer 2 (the error-driven encoder). Suppose W has bounded operator norm $\|W\|_{\text{op}} \le L_W$. Then the cluster consistency guarantee of Theorem 3 extends to the projected features $\psi(x) = W\phi(x)$ provided:

$$\Delta_{\min}(W) = \min_{i \neq j} \|W\mu_i - W\mu_j\|_2 \geq \frac{\Delta_{\min}}{L_W}$$

Proof of Corollary 2.

Step 1: Separation preservation. For any two true centers μ_i, μ_j:

$$\|W\mu_i - W\mu_j\|_2 \geq \frac{\|\mu_i - \mu_j\|_2}{\|W\|_{\text{op}}}$$

by the definition of the operator norm $\|W\|_{\text{op}} = \sup_{\|v\|=1} \|Wv\|_2$ and the fact that $\|Wv\|_2 \geq \|v\|_2 / \|W^{-1}\|_{\text{op}}$ when W is invertible on its range. More generally, the worst-case separation contraction factor is the smallest singular value of W:

$$\sigma_{\min}(W) = \inf_{\|v\|=1} \|Wv\|_2$$

Thus $\Delta_{\min}(W) \geq \sigma_{\min}(W) \cdot \Delta_{\min}$.

Step 2: Noise propagation. For the projected noise $W\varepsilon$:

$$\mathbb{E}[\exp(t \cdot u^\top W\varepsilon)] = \mathbb{E}[\exp(t \cdot (W^\top u)^\top \varepsilon)] \leq \exp(\sigma^2 \|W^\top u\|_2^2 t^2 / 2)$$

Since $\|W^\top u\|_2 \leq \|W\|_{\text{op}}$, the projected noise is sub-Gaussian with parameter $\sigma_W^2 = \sigma^2 \|W\|_{\text{op}}^2$.

Step 3: Application of Theorem 3. The projected data $\psi(x) = W\phi(x) = W\mu_{s(x)} + W\varepsilon$ satisfy the same generative model with: - Centers: $W\mu_1, \ldots, W\mu_K$ - Noise: $W\varepsilon$ with parameter $\sigma_W^2 = \sigma^2 \|W\|_{\text{op}}^2$ - Separation: $\Delta_{\min}(W) \geq \sigma_{\min}(W)\Delta_{\min}$

The separation-to-noise ratio for the projected data is:

$$\frac{\Delta_{\min}(W)^2}{\sigma_W^2} \geq \frac{\sigma_{\min}(W)^2 \Delta_{\min}^2}{\sigma^2 \|W\|_{\mathrm{op}}^2} = \frac{\Delta_{\min}^2}{\sigma^2} \cdot \kappa(W)^{-2}$$

where $\kappa(W) = \|W\|_{\mathrm{op}} / \sigma_{\min}(W)$ is the condition number of W.

If W is well-conditioned ($\kappa(W) = O(1)$), the separation-to-noise ratio is preserved up to a constant, and the same consistency guarantee holds.

Step 4: Deformation of the two-layer regime. In SCX, Layer 2 is the error-driven encoder that projects domain features using information from expert errors. The key insight is that even when cluster separation in the original ϕ-space is marginal, the error-driven projection can *increase* separation. Formally, if W is learned to maximize separability of states based on expert disagreement patterns, we may have:

$$\frac{\Delta_{\min}(W)^2}{\sigma_W^2} > \frac{\Delta_{\min}^2}{\sigma^2}$$

This is precisely the regime where two-layer SCX outperforms single-layer state discovery. Corollary 2 formalizes the condition under which this improvement does not degrade consistency.

□

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11. Corollary 3: Misclassification Rate

Corollary 3 (Exponential Tail for Misclassification). Under the conditions of Theorem 3, for any $t > 0$:

$$\mathbb{P}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{1}\{\hat{s}(x_i) \neq s(x_i)\} > t\right) \leq K \cdot \exp\left(-c \cdot \frac{nt\Delta_{\min}^2}{K}\right)$$

for a universal constant $c > 0$.

Proof Sketch. This follows from the same chaining argument as Theorem 3, combined with a refined concentration bound. For each center k, the fraction of misclassified points from that state is bounded by:

$$\mathbb{P}(\hat{\mu}_k \text{ is mis-estimated}) \leq \exp\left(-c \cdot \frac{n\Delta_{\min}^2}{K}\right)$$

when the uniform convergence bound holds with $\delta = \Theta(\Delta_{\min}^2)$. The factor K in the exponent comes from the metric entropy penalty (the need to search over K centers). Applying the union bound over K states gives the factor K outside the exponential.

□

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12. Practical Interpretation

12.1 What This Means for SCX Users

Theorem 3 provides an operational guide for SCX state discovery:

Sample requirement per state. The condition $\Delta_{\min}^2 = \omega(\sigma^2 K \log n/n)$ can be rewritten as:

$$n_k = \frac{n}{K} \geq \frac{C_0 \sigma^2 \log n}{\Delta_{\min}^2}$$

For fixed separation $\Delta_{\min}$ and noise σ^2, the minimum samples per state scales as $\log n / \Delta_{\min}^2$. Roughly:

- $\Delta_{\min}/\sigma \approx 1$ (barely separated states): need $n_k = O(\log n)$ samples per state
- $\Delta_{\min}/\sigma \approx 3$ (well separated states): need $n_k = O(\log n/9)$ samples per state, essentially a constant

Rule of thumb: For reliable state discovery, ensure at least $n_k \geq 20$ samples per state when states are well-separated ($\Delta_{\min}/\sigma \approx 3$), and $n_k \geq 100$ per state when separation is marginal ($\Delta_{\min}/\sigma \approx 1$).

12.2 The Bias-Variance Tradeoff in Practice

Regime	K too small	K too large	K optimal
Within-state variance	High (states merged)	Low (pure substates)	Balanced
Estimation error	Low (few params)	High (many params)	Balanced
Misclassification	High (merged states)	High (fragmented states)	Minimal
SCX noise detection	Fails (mixed states)	Fails (noisy estimates)	Optimal

12.3 Verifying the Conditions

Users can check whether Theorem 3 applies to their data:

1. **Estimate $\Delta^2_{\min}$**: Run SCX state discovery, compute pairwise distances between estimated centers. The minimum observed distance is a proxy for $\Delta_{\min}$ (conservative, as estimation error inflates distances).
2. **Estimate σ^2**: Within-state variance, averaged across states: $\hat{\sigma}^2 = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{n_k}\sum_{i:\hat{s}(x_i)=k}\|\phi(x_i)-\hat{\mu}_k\|_2^2$.
3. **Check separation**: Compute $\hat{\Delta}^2_{\min}/\hat{\sigma}^2$. If this exceeds $C_0 \cdot K \log n/n$, Theorem 3 guarantees consistency.

12.4 Phase Transition

There is a sharp threshold at:

$$\frac{\Delta^2_{\min}}{\sigma^2} = \Theta\left(\frac{K \log n}{n}\right)$$

- **Above threshold**: State discovery succeeds with high probability (Theorem 3 regime)
- **Below threshold**: State discovery fails, consistent with Theorem 2 (weak feature failure)

This transition is the SCX equivalent of the “Baik-Ben Arous-Peche” phase transition in spiked covariance models, or the “signal-to-noise ratio” threshold in mixture models.

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13. Connection to Theorem 2

13.1 Two Sides of the Same Coin

Aspect	Theorem 2 (Negative)	Theorem 3 (Positive)
Question	When does SCX fail?	When does SCX succeed?
Condition	ϕ is δ-weak ($I(\phi; S) \leq \delta$)	ϕ is strong ($\Delta^2_{\min} \gg \sigma^2 K \log n/n$)
Consequence	$AUC \leq AUC_{\text{base}} + O(\sqrt{\delta})$	Misclassification $\to 0$ with rate $O(\sqrt{n/(K \log n)}^{-1})$
Transition	Performance degrades continuously as $\delta \to 0$	Performance improves continuously as $\Delta_{\min} \to \infty$
Information measure	$I(\phi; S)$ (mutual information)	$\Delta^2_{\min}/\sigma^2$ (signal-to-noise ratio)

13.2 Mutual Information and Separation

The two conditions are linked. When the features follow $\phi(x) = \mu_{s(x)} + \varepsilon$ with sub-Gaussian noise, the mutual information is:

$$I(\phi(X); S) = H(S) - \sum_{k=1}^{K} \pi_k \cdot \mathbb{E}\left[\log \sum_{j=1}^{K} \frac{\pi_j}{\pi_k} \frac{p_\varepsilon(\phi - \mu_j)}{p_\varepsilon(\phi - \mu_k)} \mid S = k\right]$$

For well-separated Gaussian clusters ($\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$):

$$I(\phi(X); S) \approx \log K - \frac{1}{K} \sum_{i \neq j} \exp\left(-\frac{\|\mu_i - \mu_j\|_2^2}{4\sigma^2}\right) + o(1)$$

Thus δ (from Theorem 2) is small precisely when $\Delta^2_{\min}/\sigma^2$ is large. The weak feature condition $I(\phi; S) = \delta \to 0$ is equivalent to $\Delta^2_{\min}/\sigma^2 \to 0$, which is the condition under which Theorem 3's guarantee fails.

13.3 Unified Threshold

The two theorems together imply a critical threshold for SCX state discovery:

$$I(\phi; S) \gtrsim \frac{K \log n}{n} \cdot \log\left(\frac{\Delta^2_{\min}}{\sigma^2} \cdot \frac{n}{K \log n}\right)$$

- Above this threshold: strong features regime, Theorem 3 applies
- Below this threshold: weak features regime, Theorem 2 applies

This threshold bridges the negative result (Theorem 2) and the positive result (Theorem 3), showing that there is no gap – the two regimes are complementary.

13.4 Practical Diagnostic

For practitioners, the unified diagnostic is:

$$\frac{\hat{\Delta}^2_{\min}}{\hat{\sigma}^2} \text{ vs. } \frac{K \log n}{n}$$

Condition	Regime	SCX Expected Performance
$\hat{\Delta}^2_{\min}/\hat{\sigma}^2 \gg K \log n/n$	Strong features	State discovery reliable, SCX noise detection effective

Condition	Regime	SCX Expected Performance
$\hat{\Delta}^2_{\min}/\hat{\sigma}^2 \ll K \log n/n$	Weak features	State discovery fails, SCX degenerates to loss baseline
$\hat{\Delta}^2_{\min}/\hat{\sigma}^2 \approx K \log n/n$	Transition	Mixed performance, may need more data or stronger features

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Appendix A: Proof of the Strong Separation Lower Bound

Let $\phi(x) = \mu_{s(x)} + \varepsilon$ with $\varepsilon \sim \text{sub-Gaussian}(0, \sigma^2 I)$. We wish to bound:

$$P\left(\min_{j\neq s(x)} \|\phi(x)-\mu_j\|_2 \leq \|\phi(x)-\mu_{s(x)}\|_2\right)$$

i.e., the probability that a point from state s is closer to a different center than its own.

For any fixed center μ_j with $j \neq s(x)$:

$$\begin{aligned}\|\phi(x)-\mu_j\|_2^2-\|\phi(x)-\mu_{s(x)}\|_2^2 &= \|\mu_{s(x)}-\mu_j+\varepsilon\|_2^2-\|\varepsilon\|_2^2 \\ &= \|\mu_{s(x)}-\mu_j\|_2^2+2\varepsilon^\top(\mu_{s(x)}-\mu_j) \\ &\geq \Delta_{\min}^2-2\|\varepsilon\|_2\cdot\|\mu_{s(x)}-\mu_j\|_2\end{aligned}$$

By the sub-Gaussian tail bound, $\|\varepsilon\|_2^2 \leq \sigma^2(d_\phi+2\sqrt{d_\phi\log n}+2\log n)$ with probability at least $1-n^{-1}$. Conditioned on this event:

$$\|\phi(x)-\mu_j\|_2^2-\|\phi(x)-\mu_{s(x)}\|_2^2 \geq \Delta_{\min}^2-2M\sqrt{\sigma^2(d_\phi+2\sqrt{d_\phi\log n}+2\log n)}$$

When $\Delta_{\min}^2=\omega(\sigma^2\log n)$, the right-hand side is positive for large n, and the misclassification event has probability $O(\exp(-c\Delta_{\min}^2/\sigma^2))$.

This is the per-point separation guarantee needed in Lemma 5 Step 4.

Appendix B: Table of Key Bounds

Quantity	Bound	Source
$\log\mathcal{N}(\varepsilon,\Theta_K,\|\cdot\|_\infty)$	$Kd_\phi\log(1+2M/\varepsilon)$	Lemma 4

Quantity	Bound	Source
$\mathbb{P}(\sup \lvert W_n - W \rvert > \delta)$	$(1 + 8C_1 M/\delta)^{Kd_\phi} \exp(-n\delta^2/32C_1^2)$	Lemma 3 + 4
$W(\mu) - W(\mu^*)$	$\geq \pi_{\min}\Delta_{\min}^2$ for $\mu \neq \mu^*$	Lemma 5
Misclassification rate	$O(\sqrt{n/(K \log n)}^{-1})$ under $\Delta_{\min}^2 = \Theta(K \log n/n)$	Theorem 3
Optimal K^*	$\Theta((n/\log M)^{1/3})$	Corollary 1

Appendix C: Relationship to Other Clustering Consistency Results

Result	Setting	K scaling	Rate	Technique
Pollard (1981)	Any distribution	Fixed K	Strong consistency	Empirical process
Pollard (1982)	Any distribution	Fixed K	$n^{-1/2}$ CLT	Delta method
Bartlett et al. (1998)	Bounded distributions	Fixed K	$O(\sqrt{Kd/n})$	VC dimension
This theorem	**Sub-Gaussian mixtures**	$K = o(n^{1/3})$	$O(\sqrt{n/(K \log n)}^{-1})$	**Metric entropy + separation**

Result	Setting	K scaling	Rate	Technique
Lei et al. (2013)	General	$K = o(n)$	Minimax optimal	Generic chaining

The key novelty is the explicit handling of $K \to \infty$ with dependence on the separation-to-noise ratio, which connects clustering consistency to the SCX feature quality.

Revision Notes: - **Date**: 2026-06-27 - **Version**: 1.0 (initial draft) - **Key choices**: - The $K = o(n^{1/3})$ condition is a conservative sufficient condition derived from worst-case ($d_\phi = 1$) bias-variance tradeoff. For higher-dimensional features, K can grow faster (up to $o(n/\log n)$ when separation is maintained). - The proof uses metric entropy (covering numbers) rather than VC dimension for explicit dependence on d_ϕ and K. - The separation condition $\Delta_{\min}^2 \geq C_0 \sigma^2 K \log n / n$ is presented as an explicit inequality rather than an asymptotic statement, to facilitate practical diagnostics. - Corollary 2 (two-layer extension) shows that SCX's error-driven projection does not harm consistency when it is well-conditioned, and may improve it by increasing separation.